

Agentic-Trader — Per-Algorithm Improvement Report

Repo: <https://github.com/jeffthebever2/agentic-trader>

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Method: Six parallel read-only audits (Claude Code) covering 24 algorithm files across scoring, sizing/exits, signals/RL, ML training/grading, safety/execution, and the thematic system. No code was modified. Every finding below cites `file:line` and is tagged **SAFE FIX (do now)** — a behavior-preserving bug/coherence fix — or **EDGE (needs walk-forward first)** — anything that changes what gets traded, sized, or labeled and therefore must be validated walk-forward before shipping, one variable at a time, per the project's hard rules.

How to read this

The single most important meta-finding, repeated independently in five of the six audits, is this: **most of your remaining "profitability" problems are not missing-edge problems — they are broken-plumbing problems.** Risk controls that were written, marked "done," and trusted are silently inert: circuit breakers reading keys nothing writes, drift detectors never wired to halt, reliability trackers keyed on the wrong field, calibration that re-trains a weaker model instead of calibrating, and exit overrides captured into a variable then never used.

That is good news. It means a large fraction of the highest-value work is **safe, do-now bug fixing** that doesn't need a single walk-forward run — and it explains why the harsh leak-free reality (ML PF 0.89, deployed ROC 0.512) has been hard to move: several of the safety nets meant to catch the failure modes can't actually fire.

Nothing below proposes weakening a gate or claiming a number without evidence. The genuinely edge-changing items are flagged as such and gated on walk-forward validation.

Tier 0 — Inert safety controls (HIGH severity, mostly SAFE FIX). Fix these first.

These are controls the system *believes* it has but doesn't. Each is a Rule-2 ("breakers must actually fire") or fail-open issue. Fixing them doesn't change your edge — it stops you flying blind.

#	Algorithm / File	Finding	What's broken	Fix	
1	Thematic auto — web/api/thematic_auto.py:1878	A1	Daily-loss circuit breaker sums <code>state["closed_today"]</code> filtered on <code>close_date</code> — nothing in the codebase ever writes those keys . Closed trades live in <code>state["trades"]</code> keyed by <code>exit_time/pnl</code> , and thematic exits don't append to any trade list. So <code>realized_today</code> is always 0 and the daily-loss limit is dead code.	Read <code>state["trades"]</code> , parse <code>exit_time==today, sum pnl</code> ; make <code>_check_thematic_exits(execute=True)</code> append closed positions to that list.	SAFI (rea EDG acc
2	Safety — production_safety.py:794	E2- PS2 (DL-5)	<code>DriftDetector.check().has_drift</code> (calibration drift, high-conf failure, paper-vs-WF gap) is computed but never enters <code>check_all</code> — it only reads a scalar drift field. The most informative degradation signal can't halt.	In <code>check_all</code> , load grades, call <code>DriftDetector.check(...)</code> , append alerts to <code>halt/warn</code> .	EDG live
3	Safety — production_safety.py:3377 (caller)	E2- PS3	No FORCE_FLATTEN path . Every halt is <code>ENTRY_HALT</code> only: on a ~12% drawdown or regime-collapse halt the runner blocks new entries but never liquidates open exposure — the book keeps riding the exact condition the breaker called catastrophic.	Split halts into <code>ENTRY_HALT</code> vs <code>FORCE_FLATTEN</code> ; flatten/ tighten on drawdown / regime-collapse / ROC-broken.	EDG (arci
4	Safety — production_safety.py	E2- PS4 (DL-9)	<code>REGIME_COLLAPSE</code> (per-regime <code>WR<0.35</code>) computed by <code>reliability_stats</code> but never fed to <code>check_all</code> — a regime where the model loses 65% of the time keeps trading.	Read the reliability alert list in <code>check_all</code> ; map <code>REGIME_COLLAPSE</code> to <code>halt/warn</code> .	EDG live
5	Exits — safe_trade_guard.py:201 + caller paper_trade_today.py:3412	E2- SG1 (DL-8)	High-VIX 2-day max-hold override is built, returned, captured into <code>_hv_max_hold</code> , then never read — exits use <code>args.max_hold_days</code> . Positions opened in high-vol regimes ride the full 10–14-day clock.	Clamp per-position <code>max_hold_days</code> to <code>_hv_max_hold</code> when set, at exit-plan build (~L3598).	EDG (cha exit
6	Drift — drift_detector.py:160	DD-5	<code>_check_paper_vs_wf</code> reads <code>wf_win_rate</code> — a key the trainer never emits (it writes <code>actual_win_rate</code>). So the paper-vs-WF gap check is silently dead (always None).	Read <code>walk_forward.actual_win_rate / high_conf_win_rate</code> from the real report schema.	SAFI

#	Algorithm / File	Finding	What's broken	Fix	
7	Drift — drift_detector.py:99	DD-2	_check_calibration slices grades[-window:] with no chronological sort — grades arrive in file-glob order, so the "most recent 30" are the last 30 by filename, not by date. Every windowed drift check measures the wrong trades.	Sort grades by graded_at at the top of check().	SAFI
8	Reliability — ticker_reliability.py:67	TR-3	get_score filters trades on "pn1" in t strictly, but the grader emits pn1_pct/actual_return. If the dict schema doesn't carry pn1, every ticker scores the neutral 0.5 and the entire "stop trading losing names" tracker is silently inert.	Accept pn1/pn1_pct/return_pct/actual_return aliases; assert non-empty when trades exist.	SAFI
9	Grader — prediction_grader.py:189	PG-1 (GC-1)	actual_large_loss = max_dd_pct <= -0.08 depends on a MAE field SELL never emits → defaults 0.0 → the large-loss head always grades "correct." The one trusted signal (large_loss, ROC 0.73) has a dead grading track.	Grader: if MAE absent, set actual_large_loss=None and exclude from accuracy (surfaces the gap loudly).	SAFI (gra
10	Calibration — reliability_stats.py:57	RS-1 (GC-7)	calibration_error = abs(avg_prob - win_rate) is first-moment-only: a constant-0.6 predictor with zero discrimination scores ~0 and looks "perfectly calibrated." No monotonicity / base-rate check — exactly the blind spot that lets an anti-predictive high-confidence bucket pass.	Add a reliability-diagram check across confidence buckets: alert when high.WR ≤ low.WR or high.WR ≤ base rate.	EDGE aler hist

Why Tier 0 is first: items 1–10 are the reason the system can't see its own failures. Several are pure SAFE FIX (6, 7, 8, 9). The EDGE ones (2, 3, 4, 5, 10) wire real controls into the halt path — validate the thresholds on history, but do not weaken them.

Tier 1 — Fail-open valuation & data plumbing (HIGH, mostly SAFE FIX)

Every HIGH safety finding is the same failure mode: **uncertainty resolves to an optimistic default instead of a halt**. Each one silently neuters a control the recent V-16...V-26 work just added, because those controls consume the corrupted "looks fine" output.

#	Algorithm / File	Finding	What's broken	Fix	Class
11	State — state.py:85	E2-ST1	price = price_lookup(t) or self.entry_price. On a fetch failure/outage the position is valued at cost → unrealized PnL 0, and check_stops_and_limits continues on None → stops never fire during an outage . This is the V-25 hole, still live.	One price snapshot per cycle; on fetch failure carry last-known price, else mark the cycle uncertain (halt/skip) — never substitute entry.	SAFE FIX (hardening)
12	Safety — production_safety.py:503 + caller :568	E2-PS1	Missing/stale price values a position at entry_price → a name down 40% on a failed-quote day contributes \$0 to drawdown/loss breakers, defeating V-17 HWM and V-18 open-MTM exactly when risk is highest.	Use last-known price; if none, count the breaker as data-uncertain → halt-eligible. Never fall back to cost.	SAFE FIX (hardening)
13	State — state.py (multiple)	E2-ST2	Each property re-fetches price_lookup independently → total_value, exposure, and stop checks can use different prices within one decision .	Single snapshot at cycle start, passed into all valuation methods.	SAFE FIX
14	Fidelity — web/api/fidelity.py:1038	E2-FP5	Cash defaults to grand_total * 0.10 when the cash line fails to read — invents a balance that sizing/can_buy then trade against. (This is the cash-fallback flagged in the brief; actual constant is 0.10, not 0.15.)	On cash-read failure, hard-abort the read. A safety system must refuse to act on an unknown balance.	SAFE FIX

#	Algorithm / File	Finding	What's broken	Fix	Class
15	Fidelity – fidelity_portfolio.py:143,199	E2-FP1 / E2-FP2	A row that fails numeric parse returns <code>None</code> and is silently dropped ; an unknown account shape returns <code>[]</code> . A funded account can look flat → re-buy held names, skip stop management.	Never drop a row with a valid ticker silently; if data is non-empty but yields zero rows, fail loudly.	SAFE FIX
16	Safety – production_safety.py:356–422	E2-PS6 / E2-PS7	Four bare <code>except: pass</code> blocks turn a malformed model/drift/ROC file into "model healthy"; an f-string crash on missing WR fields silently drops the drift halt (V-22 can't evaluate).	On parse failure append a warn/halt, never pass; build the halt reason after the threshold check, format defensively.	SAFE FIX

Tier 2 — The scoring-coherence problem (one root cause, HIGH)

There is no single source of truth for the alpha score. Brain, engine, and ranker each re-implement the same formula with quietly diverging constants — so the same candidate lands in different tiers depending on which module scores it, breaking the claim that backtested (engine) edge transfers to live (brain).

Component	unified_brain	alpha_engine	candidate_ranker		
numerator	regime×breakout	regime×breakout	regime only		
reliability reward slope	0.10	0.10	0.15		
vol-penalty threshold default	0.04	0.04	0.03		
liq penalty (no ADV)	always +0.10	0 (gated)	none		
breakout clip [0,100]	no	yes	n/a		
#	Algorithm / File	Finding	What's broken	Fix	Class
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17	Brain – unified_brain.py:342	UB-A1	_dedup_ticker builds a "best-of-everything" synthetic candidate: confidence=max, ll_prob=min, tbs=max, timeout=min each from a <i>different</i> source, geometry from a third. It represents a trade no source proposed and corrupts the ll_hard_cap/min_rr gates → optimistic entries backtest can't reproduce.	Atomic source selection: pick one winning source and take ALL fields from it.	EDGE
18	All three – alpha_engine.py:342, unified_brain.py:503, candidate_ranker.py:242	AE-A5 / CR-A1	The ranker numerator is reg_score alone (constant within a scan) → intra-scan ranking is degenerate, near-zero correlation with realized edge. Brain/engine numerator is regime×breakout with no validated per-trade edge term.	Extract ONE shared scoring function used by all three (the documented Wave-4 fix); then promote calibrated 1–large_loss (the only ROC>0.5 head) into the numerator, single-counted.	EDGE (B6/B7)
19	Ranker / Engine – candidate_ranker.py:259, :115	CR-A2 / CR-A5	Reliability reward slope 0.15 (vs 0.10) and vol threshold 0.03 (vs 0.04) → ranker systematically orders candidates differently than the live engine.	Unify slope to 0.10 and vol threshold to 0.04 (confirm the paper_trade_today call site first).	SAFE FIX (unify constant)

Component	unified_brain	alpha_engine	candidate_ranker		
20	Engine — alpha_engine.py:353	AE-A3	rel_clipped is computed, unused in the math, then exported as the audit value — telemetry reports a different reliability than what scored the trade. (Doc said this was deleted; it was removed in ranker, still present here.)	Drop rel_clipped; export the real reliability used by the curve.	SAFE FIX
21	Brain — unified_brain.py:472,499,478	UB-A8 / A9 / A10	Latent config landmines: min_rr fallback 1.5 (config 1.15), breakout_max_boost fallback 0.30 (config 0.50), dead win_prob clauses that re-activate under partial-config override and revert to pre-Cycle-44 behavior.	Align fallbacks to 1.15 / 0.50; delete the dead win_prob clauses.	SAFE FIX
22	Brain — unified_brain.py:500	AE-A6	Brain doesn't clip breakout_score to [0,100] before /100 (engine does), so a malformed breakout>100 yields boost above the cap in the brain only.	Adopt engine's min(100, breakout) clip in the brain.	SAFE FIX

Tier 3 — Sizing & exit logic (mix)

The brain uses a **flat inline vol-target sizer** (unified_brain.py:704); PositionSizer's richer Kelly/ML logic only runs on the paper_trade_today.py path. ShortHoldExitManager is the live exit engine; ExitManager is **dead** (only its own docstring references it). The correlation gate is wired only in trading_graph.py:460 — **never on the live entry loops**.

#	Algorithm / File	Finding	What's broken	Fix	Class
23	Sizing — position_sizing.py:262 (+ brain :704, :724)	PS-1	ATR sizing ignores RR. A 1.2-RR and a 4-RR setup of equal stop-distance get identical share counts; expectancy $\propto$ p-RR, so the worst-expectancy bucket is over-bet. Present on both sizing paths (brain computes RR at L724 only for audit, never feeds it back). The clearest single expectancy leak.	Add a bounded RR tilt to risk_dollars: <code>rr_factor = clip(RR/rr_ref, 0.8, 1.3)</code> .	EDGE (highest sizing upside)
24	Exits — short_hold_exits.py:420	SH-1	build_exit_plan factory still carries pre-Cycle-44 fallbacks partial_profit_trigger=0.50 (should be 0.833) and min_rr=1.5 (should be 1.15). Cycle 44 claimed <i>both</i> factories were fixed; this one was missed → trained 1.2/1.15 geometry corruption for any caller.	Set defaults to 0.833 / 1.15 to match from_candidate.	SAFE FIX
25	Drawdown — drawdown.py:20	DD-1	should_keep_trading(unrealized_pnl_pct=0.0) defaults open-MTM to 0; one call site omitting it silently re-opens the "hold losers to dodge the daily halt" hole (TP-3).	Make unrealized_pnl_pct required (or self-fetched); assert at call sites.	SAFE FIX (hardening)
26	Drawdown — drawdown.py:67	DD-3	ret = pnl/100 if abs(pnl)>1 else pnl units-guess inside the breaker can misclassify a daily PnL (a \$0.8 gain treated as 80%).	Require explicit pnl_pct; remove the magnitude guess.	SAFE FIX
27	Correlation — correlation.py:28	CO-1	Correlation concentration cap is correct but dead on the live book — only called in trading_graph.py. Live book can stack arbitrarily many correlated names. (DL-3.)	Wire check_concentration_risk into the live buy decision with per-scan matrix caching (it does a live 1y yf.download per call).	EDGE (wiring is safe; enabling changes trades)
28	Exits — short_hold_exits.py:261	SH-4	MAX_HOLD is checked before STOP: a day-10 gap-down can record MAX_HOLD at current price instead of STOP_HIT at the stop.	Check stop within/before MAX_HOLD; floor the MAX_HOLD fill at the stop on a breach (tightens recorded exit only).	SAFE FIX
29	Sizing — position_sizing.py:252-279	PS-3	tier_factor is applied up to 3× on the ATR path but 1× on the fallback path → identical B-tier candidates size differently depending only on whether atr>0.	Apply tier_factor exactly once per path; unit-test equal effective tier scaling (verify A/A+ sizing unchanged).	SAFE FIX
30	Exits — exit_manager.py:1	EM-1	The dead ExitManager module's docstring asserts it is the live exit calculator → a maintainer will tune the wrong file.	Add a "NOT ON LIVE PATH — see ShortHoldExitManager" banner, or delete the module.	SAFE FIX

Tier 4 — Signals, regime & RL (mix; one real leakage)

Good news first: the screener/breakout/regime feature math has no future leakage — windows correctly use prior-bar or legitimately-known-at-entry (close) slices. The real look-ahead is in the RL environment.

#	Algorithm / File	Finding	What's broken	Fix	Class
31	Breakout + RL — breakout_scanner.py:713, rl/environment.py:354	BO-1 / RL-3	MACD histogram is always 0. Both build the signal line from a one-element <code>pd.Series([macd])</code> , so <code>signal==macd</code> → <code>hist=0</code> . Two breakout ML features (<code>macd_hist</code> , <code>macd_hist_slope3</code>) and one RL state feature are dead constants. <code>screener._macd:230</code> shows the correct full-series form.	Compute signal as 9-EMA of the MACD series, then <code>hist = (macd_series - signal_series).iloc[-1]</code> .	SAFE FIX (code) → retrain + WF (feature dist shifts)
32	RL env — rl/environment.py:218,124	RL-1	Intrabar look-ahead. The agent observes today's close and fills at today's close — the same pathology that dropped ML PF 1.19→0.89. The file's own TODO acknowledges EOD decisions should fill at next open.	Fill at <code>t+1</code> open (add <code>next_open_prices</code> to step), or shift obs to end at <code>t-1</code> .	EDGE (makes every RL eval honest)
33	Screener — screener.py:43,407	SC-1	Target-geometry mismatch — the documented #1 remaining lever. Live targets 1.2 ATR but the deployed model's labels target a ~0.75-ATR move → predicted "winners" need a bigger move than the model was trained to expect. Stop side is now aligned (1.0 ATR).	Either align live target to 0.75 ATR, or produce a clean 1.2/1.0 retrain with enough rows (Cycle 46 removed the min-price/min-adv filters; verify WF ROC≥0.49 and re-check target-hit calibration).	EDGE
34	Regime — market_regime.py:428	MR-1 / MR-4	Hard binary regime cliffs. VIX>35 or SPY 1¢ across SMA200 flips the whole book between full-size and no-trade; the "probabilistic" sub-terms are themselves <code>float(x>thr)</code> step functions.	Drive <code>no_trade/size_factor</code> off the continuous <code>crash_risk_score</code> with a smooth taper (already half-done at L562); replace the cliffs with bounded ramps.	EDGE
35	RL inference — rl/rl_signal.py:196	RS-1	Train/serve skew. Inference builds the obs by a separate code path that (a) inherits the MACD-zero bug, (b) hardcodes <code>weight=0</code> , <code>pn1=0</code> so the live agent never sees its real position, (c) zero-pads missing tickers instead of erroring. Live RL scores can be uncorrelated with the trained policy.	Build the live obs from the same function as <code>_get_obs</code> ; pass real weights; return None (skip) on missing data.	SAFE FIX (skew)
36	Breakout / Screener — breakout_scanner.py:410, screener.py:380	BO-2 / SC-2	<code>breakout_score</code> saturates at 0 for the coiling candidates the pipeline actually feeds it (volume/confirmation buckets demand a <i>same-day</i> surge) — explains the documented "breakout_score=0" candidates. The feature carries little information in production.	Re-center buckets for coil detection (reward proximity-to-resistance + dry-up), or feed the scanner an already-breaking universe; measure ROC before keeping it in the numerator.	EDGE
37	RL reward — rl/environment.py:139	RL-2	Reward = raw portfolio log-return – commission. No drawdown/vol/holding-cost term → a TD3 maximizing this just levers into whatever rose in the bull sample; not the drift/survival edge the rest of the system targets.	Add a downside-deviation / max-DD penalty and/or per-step holding cost; align with the survival edge.	EDGE

#	Algorithm / File	Finding	What's broken	Fix	Class
38	TD3 – rl/td3_agent.py:234	TD-3	ϵ -greedy with uniform random actions (unusual for TD3) + longs-only clipping means ~half of exploratory actions are 0 (no-op) → exploration biased toward cash. (Twin critics, delayed updates, Polyak, target smoothing verified correct .)	Switch to Gaussian action noise around the policy (reuse <code>policy_noise</code>).	EDGE
39	Regime – market_regime.py:235	MR-6	<code>len(spy_close) < 200</code> → "unknown" state with <code>size_factor=0.80</code> and trading still enabled – a data-truncation event drops you into 0.8× size rather than halting.	On insufficient SPY history return <code>no_trade=True</code> (or size ≤ 0.3).	SAFE FIX
40	Regime / Breakout – market_regime.py:532, breakout_scanner.py, rl/environment.py:314	MR-3 / RL-5 / RL-8	Dead <code>max_prob</code> + inconsistent probability set; meaningless <code>pn1_proxy</code> (mismatched denominators); global NumPy reseed inside <code>reset</code> .	Delete dead var / unify prob set; track real entry price for unrealized-PnL; use a local RNG Generator.	SAFE FIX

Tier 5 — ML training & calibration (HIGH; the model the gate validates is not the model you ship)

#	Algorithm / File	Finding	What's broken	Fix
41	Train – train_ml_models.py:146	TM-1	Calibration profit misuse. <code>CalibratedClassifierCV(clf, cv=None)</code> does not calibrate the profit model — with a non-frozen estimator it 5-fold re-trains <code>clf</code> on only the ~15% calibration slice. So the shipped "calibrated" win model is a weak small-sample re-fit, not the full-train model + calibration map. TM-2 hides it by scoring <code>brier_after</code> on the same rows the calibrator fit on.	<code>from sklearn.frozen import FrozenEstimator; CalibratedClassifierCV(FrozenEstimator(clf), cv=None);</code> evaluate calibration on the test set. (Verify against installed sklearn version.)
42	Train – train_ml_models.py:857 vs :668	TM-3	The WF-ROC the quality gate (≥ 0.49) reads is computed on a single RF/XGB model , but the deployed win_probability is a 0.60-XGB + 0.40-RF ensemble, separately calibrated. The gate validates a geometry that is never shipped.	Run the WF loop with the same ensemble+calibration recipe used for the bundle.
43	Stock-data train – train_ml_from_stock_data.py:645	SD-1	Dataset-build defaults to <code>--target-mult 0.9 --stop-mult 1.1</code> vs live 1.2/1.0. <code>retrain_weekly.py</code> overrides correctly, but any manual run trains the model on an <i>easier</i> (closer target, wider stop) label than reality → optimistic win-prob, anti-predictive HC bucket.	Change the two defaults to 1.2 / 1.0.
44	Stock-data train – train_ml_from_stock_data.py:645 + backtest.py:1399	SD-2	Even with correct mults, the feature target/stop are close-anchored while the label target/stop are trigger-anchored (<code>signal_high+0.05·ATR</code>). A feature/label geometry mismatch is baked into training — a deeper root of the anti-predictive HC bucket than SD-1.	Compute feature target/stop/setup_rr from the same trigger-entry the label uses.
45	Stock-data train – train_ml_from_stock_data.py:668	SD-3	The <code>SimpleNamespace</code> passed to <code>train_models</code> omits <code>calibrate/executed_weight/etc.</code> , so this entry point silently uses <code>executed_weight=20.0</code> with no CLI surface to disable — invisible 20× up-weighting on an overwhelmingly-rejected dataset.	Thread the missing args through <code>argparse</code> / pass an explicit dict.
46	Grader – prediction_grader.py:213, 185	PG-3 / PG-4	<code>win_prediction_correct</code> uses a hardcoded 0.60 threshold (live is ~0.52–0.62) and <code>actual_win = pn1 > 0</code> vs the model's trained 0.5% deadband (<code>_return > 0.005</code>) — so calibration-drift is measured against a different win definition than the model learned.	Wire the real operating threshold (param already exists); apply the 0.5% deadband in <code>actual_win</code> .

#	Algorithm / File	Finding	What's broken	Fix
47	Feature monitor – feature_monitor.py:59	FM-1	<code>_psi_one</code> returns 0.0 ("perfectly stable") when data is insufficient → a scarce-data feature passes both the deploy gate and runtime drift as definitively stable.	Return <code>None/nan</code> , tag status "insufficient" so the gate neither passes nor fails on it.
48	Feature monitor – feature_monitor.py:180	FM-4	<code>passes_gate = n_fail==0</code> ignores <code>n_missing</code> – a bundle can deploy with half its features absent.	Require a minimum fraction of <code>stable+watch</code> features to pass.
49	Ticker reliability – ticker_reliability.py:85	TR-1	Linear small-n blend: 2 losses (n=2) → score 0.40 = exactly the penalty edge, so two unlucky trades trip a size penalty on a fine ticker. No uncertainty shrinkage.	Beta-Binomial posterior mean with a Beta(5,5) prior → smooth, no cli at the penalty edge.
50	Train – train_ml_models.py:332, :199, :744	TM-5 / TM-8 / TM-10	Equal- <i>count</i> bucketing splits flat-probability mass arbitrarily (corrupts the monotonicity check); <code>recommended_threshold</code> is searched on the test set (latent holdout-tuning landmine); the ER regressor ($R^2 \approx 0.012$, gate disabled) is still trained and shipped every run.	Bucket by probability-value edges; label the threshold search "diagnostic, not for selection"; gate ER training behind a flag.

Tier 6 — Thematic scoring & honesty (mix)

#	Algorithm / File	Finding	What's broken	Fix	Class
51	Thematic – thematic_portfolio.py:514	P4	POST /thematic/trade writes straight into the shared paper account but skips the <code>PORTFOLIO_MAX_*</code> caps the auto path enforces → manual path can over-concentrate (gate-bypass).	Extract one <code>_inject_paper_position()</code> enforcing all caps + breakers, used by both endpoints.	EDGE
52	Thematic – thematic_auto.py:1928, thematic_portfolio.py:485	A4	R:R "gate" auto-widens the target to hit <code>min_rr</code> instead of rejecting/tightening – manufacturing a favorable ratio the trade never had (borderline Rule-1).	Reject sub-min-RR signals, or tighten the <i>stop</i> ; persist <code>rr_widened=True</code> + original target for honest reporting.	EDGE
53	Thematic – thematic_portfolio.py:550	P1	Manually-injected positions never set <code>entry_raw_score</code> , so the <code>entry_raw>0</code> guard fails and buzz-decay exits can never fire for them.	Set <code>entry_raw_score</code> from the latest scan score at injection.	SAFE FIX
54	Thematic – thematic_portfolio.py:54	P6	<code>_fetch_prices</code> falls back to <code>columns[0]</code> for any missing ticker → can assign one ticker's price to another (corrupts gains, stop/target, exits).	Only assign when the ticker's own column exists; never fall back to column 0.	SAFE FIX
55	Thematic – thematic_auto.py:1926	A3	<code>ticker</code> referenced before assignment in the R:R log ('ticker' in <code>dir()</code> is always false) – masks a would-be <code>NameError</code> and always logs ' ? '.	Move <code>ticker = sig["ticker"]</code> above the R:R block.	SAFE FIX
56	Thematic – thematic_portfolio.py:196	P3	Conviction weights are un-validated and double-count : theme + <code>supply_chain</code> reward the same AI-infra membership twice; momentum/entry/chase are all the same <code>chase_pct</code> counted three times; the 0–10 scale isn't actually 0–10.	Re-derive weights from <code>backtest_thematic_signals.py</code> ; collapse the theme/supply and chase double-counts.	EDGE

#	Algorithm / File	Finding	What's broken	Fix	Class
57	Thematic — thematic_auto.py:862, :1229	A5 / A6	Multi-source bonus double-counts the same news event across RSS/Google/Brave/DDG; scan-memory decay self-amplifies and can resurrect dead tickers.	Collapse news sources into one class for the bonus; force resurrected tickers to a strictly decaying score.	EDGE
58	Thematic — thematic_auto.py:1975	A10	approve_signal doesn't enforce the score/spike gate — it only attaches warnings, so a manual approve of a 1-scan spike (score 3) trades anyway.	Decide policy and enforce in approve_signal if spikes must never trade.	EDGE
59	Thematic — thematic_auto.py:1183, thematic_portfolio.py:93	A11 / P8	The shared paper-state read-modify-write isn't serialized — concurrent approvals / auto-exits can clobber each other (lost position / double-spent cash). os.replace is atomic per-file, not per-RMW.	One shared lock (file/asyncio) around every paper-state mutation.	EDGE

Recommended sequencing

Wave 1 — SAFE FIX, no walk-forward needed (do now). Stops the system flying blind and restores trained geometry. Each is behavior-preserving or strictly-more-correct:

- Tier 0: #6 DD-5, #7 DD-2, #8 TR-3, #9 PG-1
- Tier 1: #11–16 (fail-closed valuation/parsing contract)
- Tier 2: #19–22 (scoring-constant parity, drop rel_clipped, config fallbacks, breakout clip)
- Tier 3: #24 SH-1, #25 DD-1, #26 DD-3, #28 SH-4, #29 PS-3, #30 EM-1
- Tier 4: #31 BO-1/RL-3 (code fix; defer deploy of changed features to a retrain), #35 RS-1 skew, #39 MR-6, #40
- Tier 5: #41 TM-1 (API fix), #43 SD-1, #45 SD-3, #46 PG-3, #47 FM-1, #48 FM-4, #50
- Tier 6: #53 P1, #54 P6, #55 A3, plus the read-side of #51 A1

Wave 2 — wire the dead controls (EDGE; validate thresholds on history, never weaken). #2 DL-5, #3 FORCE_FLATTEN, #4 DL-9, #5 DL-8, #10 RS-1 monotonicity, #27 CO-1, exit-accounting half of #1 A1.

Wave 3 — the real edge levers (EDGE; nested/purged walk-forward, one variable at a time, log to docs/ALPHA_EVOLUTION_LOG.md). In priority order:

1. #33 SC-1 + #44 SD-2 — target-geometry mismatch (the documented #1 lever). Run the Cycle-46 retrain (scripts/retrain_weekly.py --months 84, verify n_rows≥1200, WF ROC≥0.49) and re-measure target-hit calibration.
2. #23 PS-1 — RR-aware ATR sizing (clearest expectancy leak, both sizing paths).
3. #18 B6/B7 — promote calibrated 1-large_loss into a single shared scoring numerator (after #17 UB-A1 atomic merge).
4. #32 RL-1 — next-open fills (every RL number is optimistic until fixed), then #37 RL-2 risk-aware reward, #38 TD-3.
5. #34 MR-1 — continuous regime taper.
6. Thematic: #51 P4, #52 A4, #56 P3, #57, #58, #59.

The honest bottom line

The audits did **not** find a hidden 10x edge — and per your own hard rules, no such claim could be made without leak-free walk-forward evidence, which the existing record (12 entry families, CPCV, DSR) says isn't there for the short-hold stock brain. What they found instead is that **a large share of the gap between "this should work" and PF 0.89 is mechanical**: broken calibration shipping a weaker model, a target/label geometry mismatch the model was trained against, dead-constant MACD features, and a row of safety controls that were marked done but can't fire.

Fixing Wave 1 won't move the backtest number by itself — but it makes every subsequent walk-forward result *trustworthy*, which is the precondition for the Wave 3 levers (target geometry, RR-aware sizing, single-source scoring) to show real, defensible improvement. The two validated edges you already have on record — the

leveraged-ETF vol-target momentum rotation (+34.4% CAGR test, PF 1.22, DSR 1.0) and the EMA-ribbon + Chandelier stock rule (+10–12%/yr OOS) — remain the only honest paths to materially higher returns, and none of the findings here contradict that.